

On the Traceability of Group Signatures: Uncorrupted User Must Exist

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Abstract

Group signatures (GS) are a well-known cryptographic primitive that provide both anonymity and traceability. Several implication results have been established, mainly focusing on different security levels of anonymity. For example, fully anonymous GS implies public-key encryption (PKE), and selfless anonymous GS can be constructed from one-way functions and non-interactive zero-knowledge proofs, among others. In this paper, we explore a winning condition for full traceability: an adversary must produce a valid group signature whose opening reveals an uncorrupted user. We present a generic construction of GS that is secure in the Bellare-Micciancio-Warinschi (BMW) model, except for the above condition, based solely on PKE. We emphasize that the proposed construction is highly artificial and practically meaningless because the verification algorithm always outputs 1 regardless of the input. This result suggests that the winning condition is essential for full traceability, namely that an uncorrupted user must exist.

1 Introduction

1.1 Group Signatures and Security Models

Group signatures (GS) [14] is an extension of digital signatures. Briefly, a user generates a signature on a message and the verifier checks the validity of the signature without identifying the user. A special authority called the opener can identify the signer. Two security notions are mainly considered, full anonymity and full traceability [5]. Informally, the anonymity guarantees that nobody, except the opener, can distinguish whether two signatures are generated by the same user or not (i.e., it implies unlinkability of signers). If the anonymity holds even when an adversary obtains all signing keys including the challenge users' ones, then it is called full anonymity. On the contrary, if an adversary is not allowed to obtain the signing keys of challenge users, then it is called selfless anonymity. The traceability guarantees that if a signature is valid, then the corresponding signer is always identified by the opener. After the seminal work by Bellare-Micciancio-Warinschi (BMW) [5], that gave formal definitions of full anonymity and full traceability in a static group setting (i.e., all users' signing keys are generated in the setup phase, and no new user can join the group after the setup phase), many extensions in terms of security and functionalities have been provided. To name a few, GS in a dynamic group setting [7, 27] (including authorities separation

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between the opener and the issuer), considering the opening soundness [42], revocable GSs [10, 25, 29, 30, 30, 34, 35], and fully dynamic GS [11] where users can join or leave the group after the setup phase. Bootle et al. [11] introduced the security model of fully dynamic GS. He et al. [23] further extended the model and introduced everlasting full dynamicity that allow signers to generate signatures that do not require verification with any specific epoch. As a drawback, an interactive pre-signing protocol needs to be run for each message to be signed. Thus, currently, the security model of fully dynamic GS [11] is the most enhanced one in terms of security and dynamicity, is recognized as a standard security definition in the GS area, and is instantiated by several complexity assumptions [8, 12, 32]. The basic difference between the static group model and the dynamic group model is how to generate a signing key. In the static group model (the BMW model), the group manager generates a signing key and issues it to a user. In the dynamic group model, the issuer and a user run an interactive key issuing protocol. One important fact is that the group manager knows all signing keys in the static group model whereas the issuer knows a part of signing key only in the dynamic group setting. Thus, in addition to a signing key that is known by the issuer, a user uses own secret key for generating a group signature. This setting allows us to define a security notion from an another perspective, i.e., non-frameability. Non-frameability guarantees that the rest of the group as well as the issuer (and the opener) are fully corrupt, they cannot falsely attribute a signature to an honest user who did not produce it.

1.2 Implication Results among GS and Other Cryptographic Primitives

In the BMW paper [5], a generic construction of GS secure in the BMW model was given from public key encryption (PKE), signatures, and non-interactive zero knowledge (NIZK) proofs. A generic construction of GS secure in the Bellare-Shi-Zhang (BSZ) model from the same building blocks was proposed [7] and it was shown that the generic construction provides the opening soundness [42]. As the opposite direction, Abdalla and Warinschi [1] showed that fully anonymous GS implies PKE. Intuitively, a public key consists of $(\text{gpk}, \text{gsigk}_1, \text{gsigk}_2)$ where gpk is a group public key and $(\text{gsigk}_1, \text{gsigk}_2)$ are two signing keys. A ciphertext of a plaintext $m \in \{0, 1\}$ is a group signature generated by gsigk_{m+1} . The opening key ok is set as a decryption key since it determines which signing key was used for generating a group signature. The anonymity hides information of which signing key was used for generating the signature. Moreover, due to full anonymity, the anonymity holds even if signing keys $(\text{gsigk}_1, \text{gsigk}_2)$ are publicly opened. This construction methodology, revealing signing keys as a public key, was employed to show other implications. Emura et al. [18] showed that GS secure in the BSZ model (with the opening soundness) implies public key encryption with non-interactive opening (PKENO) [16] and Sakai et al. [40] showed that GS with message-dependent opening (GS-MDO) implies identity-based encryption (IBE). Imura et al. [24] showed that GS-MDO directly implies timed-release encryption (TRE). Sakai et al. [41] utilized Nakanishi et al.'s revocable GS scheme [34] to construct a PKE scheme allowing a message space restriction. Here, revocation in GS allows us to revoke signing keys. Intuitively revoking a signing key in GS means that restricting a message space in the GS-based PKE context since $(\text{gsigk}_1, \text{gsigk}_2)$ specifies the message space $\{0, 1\}$.

The implication result given by Abdalla and Warinschi [1] demonstrates that PKE is an essential building block for constructing fully anonymous GS.¹ In other words, GS can be constructed from weaker primitives if full anonymity is not required. Camenisch and Groth [13] showed that selfless anonymous GS can be constructed from one-way functions (OWF) and NIZK. Katsumata and Yamada [26] showed that NIZK is not necessary to construct selfless anonymous GS.² Ohtake et

¹Bellare and Fuchsbaauer [4] showed that GS can be constructed from policy-based signatures and PKE.

²They gave a generic construction of selfless anonymous GS from indexed attribute-based signatures (ABS).

al. [36] showed that if unlinkability (which is implied by full anonymity) is not required, then GS can be constructed from identity-based signatures (which can be constructed from the standard signatures [6], and thus from OWF [39]). To sum up, whether the anonymity holds or not when signing keys are exposed is the essential condition to separate the anonymity.

1.3 Full Traceability, Revisited

From now on, we focus on the BMW model in this paper with the following three reasons. First, it contains all essential security and functionalities of GS, full anonymity and full traceability, and is one of the most widely accepted definition, e.g., [9, 19, 22, 26, 28, 31]. Second, the dynamic aspects of the group signature scheme are actually not required to construct public key encryption primitives from GS [1, 18, 40]. For example, the GS-based PKENO construction [18] employed the BSZ model which considers a dynamic group setting, the key generation algorithm of PKENO internally runs the interactive join protocol by himself. Third, non-frameability is also not required to construct public key encryption primitives from GS [1, 18, 40]. More precisely, a user i has a group signing key gsigk_i in the model BMW and i additionally has a user secret key usk_i in the BSZ model. To generate a group signature, gsigk_i is sufficient in the BMW model whereas both usk_i and gsigk_i are required in the BSZ model. To construct a PKE scheme from GS, all keys for generating a group signature need to be set as a public key. In the BMW model, $(\text{gsigk}_1, \text{gsigk}_2)$ is sufficient. In the BSZ model, $(\text{gsigk}_1, \text{usk}_1, \text{gsigk}_2, \text{usk}_2)$ needs to be made public. In the security model of non-frameability, an adversary does not know usk of the targeted honest user (though the adversary knows the corresponding gsigk since the adversary can corrupt the issuer). Thus there is no way to employ non-frameability to construct PKE since all signing keys “including user secret keys” need to be set as a public key. Of course, we do not deny any possibility to construct a cryptographic primitive including an interactive procedure from GS and then it may require the dynamic group setting as an essential condition.

So far, the above mentioned implication results focused on the (several security levels of) anonymity, and did not focus on the traceability. One exception is the GS-based PKENO construction [18]: If an untraceable but a valid signature is contained in a prove query,³ then the prove oracle cannot respond the query since it is not publicly detectable. The traceability is employed to exclude the case. As a well-known folklore, a GS scheme without the opening functionality (i.e., without traceability) can be constructed by sharing a signing key of a signature scheme among all users. That is, for a verification and signing key pair of a signature scheme $(\text{verk}, \text{sigk}) \leftarrow \text{Sig.KeyGen}(1^\lambda)$, set $\text{gpk} = \text{verk}$, and $\text{gsigk}_i = (\text{sigk}, i)$ for $i \in [n]$. Then, full anonymity holds since $\sigma \leftarrow \text{Sig.Sign}(\text{sigk}, M)$ does not contain information of i (here, M is an arbitrary signed message). However, this folklore and the GS-based PKENO construction [18] do not answer the following question:

What is the essential condition of full traceability?

We explore this research question by analyzing the winning conditions, separately.

Informally, three winning conditions are considered in the definition of full traceability in the BMW model [5]. Let n be the number of users. gpk and ok are given to an adversary $\mathcal{A}$. Moreover,

Though it can be instantiated from the standard LWE and SIS assumptions, precisely, it is not clear whether indexed ABS is weaker than PKE or not, to the best of our knowledge.

³In the syntax of PKENO, the Prove algorithm, that takes a secret key and a ciphertext, outputs a proof π . The verification algorithm, that takes a ciphertext, a plaintext (which may be $\perp$), and a proof, outputs 1 if the decryption result of the ciphertext is the plaintext. In their GS-based PKENO construction, ok is set as π when a ciphertext is valid but the decryption result is $\perp$. Because the reduction algorithm does not know ok , the traceability is employed to exclude the case, and is not required to respond a decryption query.

$\mathcal{A}$ is allowed to issue corrupt queries $i \in [n]$ that obtains gsig_i where $[n] = \{1, 2, \dots, n\}$. $\mathcal{A}$ is also allowed to issue signing queries (i, M) where $i \in [n]$. Finally $\mathcal{A}$ outputs a pair of a group signature and a message (Σ^*, M^*) . Let i be the opening result of (Σ^*, M^*) . It is required that (Σ^*, M^*) is accepted by the verification algorithm. $\mathcal{A}$ wins if one of the following three conditions holds.

1. $i = \perp$.
2. $i \neq \perp \wedge i \notin [n]$.
3. $i \in [n]$, i is not corrupted by $\mathcal{A}$, and (i, M^*) is not sent as a signing query.

Due to the definition, full traceability ensures that a valid group signature is always opened correctly. More clearly, since an adversary colludes with some users, it ensures that no colluding set of users can produce group signatures that cannot be opened, or group signatures that cannot be traced back to some user of the coalition.

1.4 Our Contribution

In this paper, we focus on the last winning condition of full traceability: an adversary is required to produce a valid group signature whose opening result is an uncorrupted user. We demonstrate that a generic construction of GS secure in the BMW model *except the last winning condition of full traceability* from PKE only.

- Especially, the construction provides full anonymity (in the CPA-setting) without NIZK whereas previous NIZK-free constructions (except the above folklore) considered a weaker notion of anonymity, selfless anonymity [26] or linkability/pseudonymity [36]. This result shows that the last winning condition is essential in full traceability.
 - The main observation here is the first and second winning conditions can be considered in the opening algorithm regardless of the behavior of the adversary whereas the last condition depends on the behavior of the adversary, i.e., who will be corrupted by the adversary.
 - We emphasize that the proposed construction is quite artificial and meaningless in practice. Our primary assertion is that, in the absence of the final condition of full traceability, GS can be constructed with remarkable ease, albeit in an artificial manner, even without NIZK. Given that a NIZK-free constructions [26, 36] constitute a technically significant contribution, we wished to emphasize the ease of construction in this context. It is not our intention to claim that the artificial construction is superior by comparison.
 - An adversary with full anonymity in the BMW model is allowed to issue opening queries, and this notion is referred to as full CCA anonymity. Our construction provides full CPA anonymity, where an adversary is not allowed to issue any opening query. We emphasize that this limitation does not hinder us from identifying what the essential condition of full traceability is.
- As a by-product of our result, we pointed out that the last winning condition does NOT appear when simply the Abdalla and Warinschi construction [1] is employed to construct a PKE scheme because all signing keys are revealed as a public key (i.e., under full corruption). Due to our first result, the underlying GS scheme can be realized via the artificial construction.

As a folklore, a GS-based PKE scheme provides public verifiability since a ciphertext is a group signature: Ohtake et al. [36] mentioned that “*It remarks that the PKE scheme, which is constructed with the above method, has public verifiability. Namely, anyone can verify validity of a ciphertext by using gpk.*”. Thus, we also discuss what public verifiability can be realized when the underlying GS scheme provides full traceability. We also pointed out that there is an artificial construction that satisfies the definitions of public verifiability introduced by Nieto et al. [21] and Faonio et al. [20].

2 Preliminaries: PKE

In this section, we define PKE which is a building block of the proposed construction given in Section 4. Let $\text{PKE} = (\text{PKE.KeyGen}, \text{PKE.Enc}, \text{PKE.Dec})$ be a PKE scheme.

Definition 1 (Syntax of PKE).

PKE.KeyGen: *The key generation algorithm takes as a security parameter λ as input, and outputs a public key and decryption key pair (pk, dk) . pk contains a plaintext space which we denote $\text{MessageSpace}(\text{pk})$.*

PKE.Enc: *The encryption algorithm takes pk and a plaintext $m \in \text{MessageSpace}(\text{pk})$, and outputs a ciphertext C .*

PKE.Dec: *The decryption algorithm takes dk and C as input, and outputs m or $\perp$.*

Correctness: We say that a PKE scheme is correct if for all $\lambda \in \mathbb{N}$, all $(\text{pk}, \text{dk}) \leftarrow \text{PKE.KeyGen}(1^\lambda)$, all $m \in \text{MessageSpace}(\text{pk})$, $\text{PKE.Dec}(\text{dk}, \text{PKE.Enc}(\text{pk}, m)) = m$ holds with overwhelming probability.

The IND-CCA security is defined as follows.

Definition 2 (IND-CCA). *Let $\mathcal{C}$ be the challenger and $\mathcal{A}$ be an adversary. We define the security model via the security game between $\mathcal{C}$ and $\mathcal{A}$. At the beginning of the game, $\mathcal{C}$ runs $(\text{pk}, \text{dk}) \leftarrow \text{PKE.KeyGen}(1^\lambda)$, and gives pk to $\mathcal{A}$. $\mathcal{A}$ is allowed to issue decryption queries.*

Decryption Query: *$\mathcal{A}$ sends a ciphertext C . $\mathcal{C}$ returns the result of $\text{PKE.Dec}(\text{dk}, C)$.*

$\mathcal{A}$ declares two equal-length plaintexts $m_0^, m_1^* \in \text{MessageSpace}(\text{pk})$. $\mathcal{C}$ flips $b \xleftarrow{\$} \{0, 1\}$, computes the challenge ciphertext $C^* \leftarrow \text{PKE.Enc}(\text{pk}, m_b^*)$, and sends C^* to $\mathcal{A}$.*

$\mathcal{A}$ is further allowed to issue decryption queries.

Decryption Query: *$\mathcal{A}$ sends a ciphertext $C \neq C^*$. $\mathcal{C}$ returns the result of $\text{PKE.Dec}(\text{dk}, C)$.*

Finally, $\mathcal{A}$ outputs $b' \in \{0, 1\}$. We say that a PKE scheme is IND-CCA secure if $|\Pr[b = b'] - 1/2|$ is negligible in λ for any PPT adversaries $\mathcal{A}$.

The IND-CPA security is defined when $\mathcal{A}$ is not allowed to issue a decryption query.

3 GS and the BMW Model

In this section, we introduce the BMW model [5]. Let $\text{GS} = (\text{Setup}, \text{GSign}, \text{GVerify}, \text{Open})$ be a GS scheme defined as follows.

Definition 3 (Syntax of GS).

Setup: The setup algorithm takes a security parameter λ and the number of group users n , and outputs a group public key gpk , an opening key ok , and signing keys $\{\text{gsigk}_i\}_{i \in [n]}$.

GSign: The signing algorithm takes gsigk_i and a message M , and outputs a group signature Σ .

GVerify: The verification algorithm takes gpk , Σ , and M as input, and outputs 0 or 1.

Open: The opening algorithm takes ok , Σ , and M , and outputs an index i or a failure symbol $\perp$.

Correctness: For all $\lambda, n \in \mathbb{N}$, all $(\text{gpk}, \text{ok}, \{\text{gsigk}_i\}_{i \in [n]}) \leftarrow \text{Setup}(1^\lambda, 1^n)$, all $i \in [n]$, and all $M \in \{0, 1\}^*$, $\text{GVerify}(\text{gpk}, \text{GSign}(\text{gsigk}_i, M), M) = 1$ and $\text{Open}(\text{ok}, \text{GSign}(\text{gsigk}_i, M), M) = i$ hold with overwhelming probability.

Next, we define full anonymity. It guarantees that an adversary $\mathcal{A}$, who does not possess ok , cannot distinguish which signing key is used for generating a group signature, even $\mathcal{A}$ has all signing keys.

Definition 4 (Full Anonymity). Let $\mathcal{C}$ be the challenger and $\mathcal{A}$ be an adversary. We define the security model via the security game between $\mathcal{C}$ and $\mathcal{A}$. At the beginning of the game, $\mathcal{C}$ runs $(\text{gpk}, \text{ok}, \{\text{gsigk}_i\}_{i \in [n]}) \leftarrow \text{Setup}(1^\lambda, 1^n)$ and gives $(\text{gpk}, \{\text{gsigk}_i\}_{i \in [n]})$ to $\mathcal{A}$. $\mathcal{A}$ is allowed to issue opening queries.

Opening Query: $\mathcal{A}$ sends (Σ, M) . $\mathcal{C}$ returns the result of $\text{Open}(\text{ok}, \Sigma, M)$.

$\mathcal{A}$ declares (i_0^*, i_1^*, M^*) where $i_0^*, i_1^* \in [n]$. $\mathcal{C}$ flips $b \xleftarrow{\$} \{0, 1\}$, computes $\Sigma^* \leftarrow \text{GSign}(\text{gsigk}_{i_b^*}, M^*)$, and gives Σ^* to $\mathcal{A}$. $\mathcal{A}$ is further allowed to issue opening queries.

Opening Query: $\mathcal{A}$ sends $(\Sigma, M) \neq (\Sigma^*, M^*)$. $\mathcal{C}$ returns the result of $\text{Open}(\text{ok}, \Sigma, M)$.

Finally, $\mathcal{A}$ outputs $b' \in \{0, 1\}$. We say that the GS scheme is fully anonymous if the advantage

$$\text{Adv}_{\text{GS}, \mathcal{A}}^{\text{full-anon}}(\lambda, n) = |\Pr[b = b'] - 1/2|$$

is negligible in λ for any PPT adversaries $\mathcal{A}$.

The above definition is also called full CCA anonymity since $\mathcal{A}$ is allowed to issue opening queries. We can consider a weaker version called full CPA anonymity where an adversary is not allowed to issue an opening query.

Next, we define full traceability. It guarantees that an adversary $\mathcal{A}$ cannot produce a valid group signature whose opening result is either $\perp$ or an uncorrupted user. We note that full traceability implies unforgeability as mentioned in the BMW paper [5]. In the original definition, the second winning condition below (i.e., the opening result is $i \notin [n]$) is implicitly contained in other conditions. However, it seems ambiguous which condition contains the second condition. For example, we can define that the **Open** algorithm outputs $\perp$ if the opening result is $i \notin [n]$, or we can also

regard that a user $i \notin [n]$ is not corrupted since $\mathcal{A}$ is allowed to issue a corrupt query for $i \in [n]$. In order to avoid any confusion, we separately define the condition below. We also remark that we can define full traceability by following strong unforgeability. Nevertheless, we follow the original BMW model where (i, M^*) is not sent as a signing query.

Definition 5 (Full Traceability). *Let $\mathcal{C}$ be the challenger and $\mathcal{A}$ be an adversary. We define the security model via the security game between $\mathcal{C}$ and $\mathcal{A}$. At the beginning of the game, $\mathcal{C}$ runs $(\text{gpk}, \text{ok}, \{\text{gsigk}_i\}_{i \in [n]}) \leftarrow \text{Setup}(1^\lambda, 1^n)$ and gives (gpk, ok) to $\mathcal{A}$. $\mathcal{C}$ initialize $\text{CorrU} = \emptyset$. $\mathcal{A}$ is allowed to issue corruption and signing queries.*

Corrupt User Query: $\mathcal{A}$ sends $i \in [n]$. Then $\mathcal{C}$ updates $\text{CorrU} = \text{CorrU} \cup \{i\}$ and returns gsigk_i to $\mathcal{A}$.

Signing Query: $\mathcal{A}$ sends $i \in [n]$ and M . Then, $\mathcal{C}$ runs $\Sigma \leftarrow \text{GSign}(\text{gsigk}_i, M)$ and returns Σ to $\mathcal{A}$.

Finally, $\mathcal{A}$ outputs (Σ^*, M^*) . $\mathcal{C}$ outputs 0 if $\text{GVerify}(\text{gpk}, \Sigma^*, M^*) = 0$. Otherwise, $\mathcal{C}$ runs $i \leftarrow \text{Open}(\text{ok}, \Sigma^*, M^*)$. $\mathcal{C}$ outputs 1 if one of the followings hold.

1. $i = \perp$.
2. $i \neq \perp \wedge i \notin [n]$.
3. $i \in [n]$, $i \notin \text{CorrU}$, and (i, M^*) is not sent as a signing query.

We say that the GS scheme is fully traceable if the advantage

$$\text{Adv}_{\text{GS}, \mathcal{A}}^{\text{full-trace}}(\lambda, n) = \Pr[\mathcal{C} \rightarrow 1]$$

is negligible in λ for any PPT adversaries $\mathcal{A}$.

We clarify that the last condition “ $i \in [n]$ and $i \notin \text{CorrU}$ ” indicates that $\mathcal{A}$ is required to produce a valid group signature whose opening result is an uncorrupted user.

4 Essential Condition of Full Traceability

In this section, we show that the last winning condition of full traceability: an adversary is required to produce a valid group signature whose opening result is an uncorrupted user, is essential. To show this, we construct a GS scheme secure in the BMW model except the last winning condition of full traceability from PKE only.

We construct a GS scheme from $\text{PKE} = (\text{PKE.KeyGen}, \text{PKE.Enc}, \text{PKE.Dec})$ as follows. As mentioned in the introduction, the construction is quite artificial and meaningless in practice because the GVerify algorithm always outputs 1 regardless of the input. In other words, GS is meaningless in practice if we do not consider the last winning condition of full traceability. This result suggests us the last winning condition is essential in full traceability.

Setup $(1^\lambda, 1^n)$: Run $(\text{pk}, \text{dk}) \leftarrow \text{PKE.KeyGen}(1^\lambda)$ and choose $i' \xleftarrow{\$} [n]$. Output $\text{gpk} = \text{pk}$, $\text{ok} = (\text{dk}, i')$, and $\text{gsigk}_i = i$ for $i \in [n]$. Here, $\text{MessageSpace}(\text{pk}) = [n]$.

GSign (gsigk_i, M) : Parse $\text{gsigk}_i = i$. Compute $C \leftarrow \text{PKE.Enc}(\text{pk}, i)$ and output $\Sigma = C$.

$\text{GVerify}(\text{gpk}, \Sigma, M)$: Output 1.

$\text{Open}(\text{ok}, \Sigma, M)$: Parse $\text{ok} = (\text{dk}, i')$ and $\Sigma = C$. Run $i \leftarrow \text{PKE.Dec}(\text{dk}, C)$. If $i \in [n]$, output i . Otherwise, if $i = \perp$ or $i \notin [n]$, then output i' .

Theorem 1. *The proposed construction is correct if the underlying PKE scheme is correct.*

Proof. Since the GVerify algorithm always outputs 1, $\text{GVerify}(\text{gpk}, \text{GSign}(\text{gsig}_i, M), M) = 1$ holds for all $i \in [n]$ and $M \in \{0, 1\}^*$. We need to show that $\text{Open}(\text{ok}, \text{GSign}(\text{gsig}_i, M), M) = i$ holds for all $i \in [n]$ and $M \in \{0, 1\}^*$. Now, an honestly generated group signature is represented as $\Sigma = C$ where $C \leftarrow \text{PKE.Enc}(\text{pk}, i)$. Since the underlying PKE scheme is assumed to provide correctness, for $i' \leftarrow \text{PKE.Dec}(\text{dk}, C)$, $i' = i$ holds with overwhelming probability. Thus, the Open algorithm correctly outputs i with overwhelming probability. This concludes the proof. $\square$

Theorem 2. *The proposed construction is fully CPA anonymous if the underlying PKE scheme is IND-CPA secure.⁴*

Proof. Let $\mathcal{C}$ be the challenger of the IND-CPA security and $\mathcal{A}$ be an adversary of full anonymity. We construct an algorithm $\mathcal{B}$ that breaks the IND-CPA security as follows. At the beginning of the game, $\mathcal{C}$ runs $(\text{pk}, \text{dk}) \leftarrow \text{PKE.KeyGen}(1^\lambda)$ and gives pk to $\mathcal{B}$. $\mathcal{B}$ sets $\text{gpk} = \text{pk}$, and sends $(\text{gpk}, \{i\}_{i \in [n]})$ to $\mathcal{A}$. $\mathcal{A}$ declares (i_0^*, i_1^*, M^*) where $i_0^*, i_1^* \in [n]$. $\mathcal{B}$ sets $(m_0^*, m_1^*) = (i_0^*, i_1^*)$ and sends (m_0^*, m_1^*) to $\mathcal{C}$. $\mathcal{C}$ returns the challenge ciphertext $C^* \leftarrow \text{PKE.Enc}(\text{pk}, m_b^*)$ where $b \in \{0, 1\}$. $\mathcal{B}$ sets $\Sigma^* = C^*$ and gives Σ^* to $\mathcal{A}$. Finally, $\mathcal{A}$ outputs $b' \in \{0, 1\}$. $\mathcal{B}$ outputs the same b' and breaks full anonymity at least the advantage of $\mathcal{A}$. $\square$

Theorem 3. *The proposed construction is fully traceable without the last condition.*

Proof. Since the Open algorithm never outputs $\perp$ or $i \notin [n]$, the advantage $\text{Adv}_{\text{GS}, \mathcal{A}}^{\text{full-trace}}(\lambda, n) = 0$. $\square$

Remark 1. The proof of Theorem 3 relies on the fact that no group signature exists in the construction for which the opening result is $\perp$. To exclude the case, we can add the condition “there exists a group signature for which the opening result is $\perp$ ”. However, this is not sufficient to bypass the artificial construction. For example, $\text{GVerify}(\text{gpk}, \Sigma, M)$ outputs 0 only when $(\Sigma, M) = (\perp, \perp)$, and $\text{Open}(\text{ok}, \Sigma, M)$ outputs $\perp$ only when $(\Sigma, M) = (\perp, \perp)$. Since $\mathcal{A}$ is required to produce a valid group signature in the definition of full traceability, $\mathcal{A}$ cannot output $(\Sigma^*, M^*) = (\perp, \perp)$ and thus $\text{Adv}_{\text{GS}, \mathcal{A}}^{\text{full-trace}}(\lambda, n) = 0$ holds.

One may think that the construction is not unforgeable when a homomorphic PKE scheme is employed. For example, let an adversary $\mathcal{A}$ modify $\text{PKE.Enc}(\text{pk}, j)$ to $\text{PKE.Enc}(\text{pk}, i)$ by homomorphically adding $\text{PKE.Enc}(\text{pk}, k)$ where $i = j + k$. Again, the winning condition is either $i = \perp$ or $i \notin [n]$. If $i \notin [n]$, then, the Open algorithm outputs $i' \in [n]$ and $\mathcal{A}$ loses. If $i \in [n]$, then it does not satisfy the winning condition above. We note that this case is captured by the last condition: $i \in [n]$ and i is not corrupted. Thus, our construction is fully traceable if the last condition is removed.

⁴Remark that the construction is not fully CCA anonymous even if the underlying PKE scheme is IND-CCA secure. Briefly, an adversary is allowed to issue an opening query (Σ^*, M) where $\Sigma^* = C^*$ is the challenge ciphertext of the PKE scheme and $M \neq M^*$. Since the GVerify algorithm always outputs 1, the reduction algorithm needs to respond the query but the reduction algorithm does not send C^* as a decryption query.

Remark 2. Our analysis shows that the last condition “ $i \in [n]$ and $i \notin \text{CorrU}$ ” is essential in the BMW model. We remark that an adversary of traceability is allowed to corrupt all users in the BSZ model [7]. We clarify that this does not contradict our result because of the Judge algorithm. That is, an adversary of traceability in the BSZ model is required to produce a valid group signature Σ and let (i, π) be the opening result where π is the proof of opening. The adversary wins if the Judge algorithm with (i, π, Σ) outputs 0 meaning that Σ is not generated by the user i . Because the opening procedure is honestly done according to the Open algorithm, traceability is well-defined even under the full corruption. Due to our observation, it is expected that there is another essential condition of traceability in the BSZ model. Exploring what the essential condition of traceability is in the BSZ model is future work of this paper.

5 Discussion: Public Verifiability of (GS-based) PKE

In this section, we discuss public verifiability of PKE constructed by GS via the Abdalla-Warinschi methodology [1] where signing keys of GS are set as a public key of PKE. Due to our observation here, we pointed out that previous definitions of public verifiability [20, 21] can be bypassed by the artificial construction.

5.1 On Public Verifiability of GS-based PKE

First, we construct a (1-bit) PKE scheme⁵ from $\text{GS} = (\text{Setup}, \text{GSign}, \text{GVerify}, \text{Open})$ as follows.

PKE.KeyGen(1^λ): Run $(\text{gpk}, \text{ok}, \text{gsigk}_1, \text{gsigk}_2) \leftarrow \text{Setup}(1^\lambda, 1^2)$ and output $\text{pk} = (\text{gpk}, \text{gsigk}_1, \text{gsigk}_2)$ and $\text{dk} = \text{ok}$.

PKE.Enc(pk, m): Parse $\text{pk} = (\text{gpk}, \text{gsigk}_1, \text{gsigk}_2)$. For $m \in \text{MessageSpace}(\text{pk}) = \{0, 1\}$, choose $M \in \{0, 1\}^*$ and run $\Sigma \leftarrow \text{GSign}(\text{gsigk}_{m+1}, M)$. Output $C = (\Sigma, M)$.

PKE.Dec(dk, C): Parse $\text{dk} = \text{ok}$ and $C = (\Sigma, M)$. Run $i \leftarrow \text{Open}(\text{ok}, \Sigma, M)$. If $i = \perp$, then output $\perp$. Otherwise, output $m = i - 1$.

The following theorems are straightforward, but we explicitly give them to clarify that full traceability is not necessary to provide the correctness and the IND-CCA security.

Theorem 4. *The PKE scheme is correct if the underlying GS scheme is correct.*

Proof. For an honestly generated group signature $\Sigma \leftarrow \text{GSign}(\text{gsigk}_{m+1}, M)$ and $i \leftarrow \text{Open}(\text{ok}, \Sigma, M)$, $i = m + 1$ holds due to the correctness of GS. $\square$

Theorem 5. *The PKE scheme is IND-CCA secure if the underlying GS scheme is fully CCA anonymous.*

Proof. Let $\mathcal{C}$ be the challenger of full CCA anonymity and $\mathcal{A}$ be an adversary of the IND-CCA security. We construct an algorithm $\mathcal{B}$ that breaks full CCA anonymity as follows. At the beginning of the game, $\mathcal{C}$ runs $(\text{gpk}, \text{ok}, \text{gsigk}_1, \text{gsigk}_2) \leftarrow \text{Setup}(1^\lambda, 1^2)$ and sends $(\text{gpk}, \text{gsigk}_1, \text{gsigk}_2)$ to $\mathcal{B}$. $\mathcal{B}$

⁵To amplify the message space, we can use M (a signed message) as a tag/label where for a key pair of a one-time signature scheme $(\text{verk}, \text{sigk})$, compute $\Sigma_i \leftarrow \text{GSign}(\text{gsigk}_{m_i+1}, \text{verk})$ for encrypting the i -th bit of the plaintext m_i , and a ciphertext is $(\sigma_{\text{OTS}}, \text{verk}, \{\Sigma_i\})$. This message space amplification technique was introduced in [18].

sets $\text{pk} = (\text{gpk}, \text{gsigk}_1, \text{gsigk}_2)$ and sends pk to $\mathcal{A}$. For a decryption query $C = (\Sigma, M)$ issued by $\mathcal{A}$, $\mathcal{B}$ forwards (Σ, M) to $\mathcal{C}$ as an opening query. If $\mathcal{C}$ returns $\perp$, then $\mathcal{B}$ returns $\perp$ to $\mathcal{A}$. Otherwise, when $\mathcal{C}$ returns $i \in \{1, 2\}$, then $\mathcal{B}$ returns $i - 1$ to $\mathcal{A}$. $\mathcal{A}$ declares (m_0^*, m_1^*) . Without loss of generality, we set $(m_0^*, m_1^*) = (0, 1)$. $\mathcal{B}$ randomly chooses $M^* \in \{0, 1\}^*$ and sends $(1, 2, M^*)$ to $\mathcal{C}$ as the challenge query. $\mathcal{C}$ returns $\Sigma^* \leftarrow \text{GSign}(\text{gsigk}_{b+1}, M^*)$ to $\mathcal{B}$ where $b \in \{0, 1\}$. $\mathcal{B}$ sets $C^* = (\Sigma^*, M^*)$ and sends C^* to $\mathcal{A}$. For a decryption query $C = (\Sigma, M)$ issued by $\mathcal{A}$, $\mathcal{B}$ responds the query as in the previous phase. We note that $\mathcal{B}$ can forward any query since $C \neq C^*$, $(\Sigma, M) \neq (\Sigma^*, M^*)$ holds. Finally, $\mathcal{A}$ outputs $b' \in \{0, 1\}$. $\mathcal{B}$ outputs the same b' and breaks full anonymity at least the same advantage of $\mathcal{A}$. $\square$

The following theorem also holds (we omit the proof because it can be obtained by the proof of Theorem 5).

Theorem 6. *The PKE scheme is IND-CPA secure if the underlying GS scheme is fully CPA anonymous.*

GS-based PKE does not require Full Traceability: Since $C = (\Sigma, M)$ in the GS-based PKE scheme given in Section 5.1, the validity of C can be publicly checked via $\text{GVerify}(\text{gpk}, \Sigma, M)$. Here, the **Open** algorithm is internally run in the **PKE.Dec** algorithm. Thus, intuitively the verifiability guarantees that if $\text{GVerify}(\text{gpk}, \Sigma, M) = 1$ for $C = (\Sigma, M)$, then $m \in \text{MessageSpace}(\text{pk})$ holds for $m \leftarrow \text{PKE.Dec}(\text{dk}, C)$. However, to reduce full traceability, the reduction algorithm issues corruption queries 1 and 2, to obtain $(\text{gsigk}_1, \text{gsigk}_2)$. Under the full corruption, the last condition of full traceability does not happen. That is, to construct an IND-CPA secure GS-based PKE scheme, it is sufficient that the underlying GS scheme provides the correctness, full CPA anonymity, and full traceability without the last condition that is exactly artificially constructed in Section 4. Since such a GS scheme can be constructed from PKE only and is meaningless in practice, the above intuition does not capture what we can achieve when a publicly verifiable PKE scheme is constructed from GS secure in the BMW model.

Here, we discuss what public verifiability can be achieved if the underlying GS scheme provides full traceability. First, we set $n > 2$ that allows an adversary to produce a valid group signature whose opening result is an uncorrupted user. For example,

PKE.KeyGen(1^λ): Run $(\text{gpk}, \text{ok}, \{\text{gsigk}_i\}_{i \in [n]}) \leftarrow \text{Setup}(1^\lambda, 1^n)$ for $n > 2$. Pick $\text{gsigk}, \text{gsigk}' \xleftarrow{\$} \{\text{gsigk}_i\}_{i \in [n]}$ and set $(\text{gsigk}_1, \text{gsigk}_2) = (\text{gsigk}, \text{gsigk}')$. Output $\text{pk} = (\text{gpk}, \text{gsigk}_1, \text{gsigk}_2)$ and $\text{dk} = (\text{gpk}, \text{ok})$.

PKE.Enc(pk, m): Parse $\text{pk} = (\text{gpk}, \text{gsigk}_1, \text{gsigk}_2)$. For $m \in \text{MessageSpace}(\text{pk}) = \{0, 1\}$, choose $M \in \{0, 1\}^*$ and run $\Sigma \leftarrow \text{GSign}(\text{gsigk}_{m+1}, M)$. Output $C = (\Sigma, M)$.

PKE.Dec(dk, C): Parse $\text{dk} = (\text{gpk}, \text{ok})$ and $C = (\Sigma, M)$. Output $\perp$ if $\text{GVerify}(\text{gpk}, \Sigma, M) = 0$. Otherwise, run $i \leftarrow \text{Open}(\text{ok}, \Sigma, M)$. If $i = \perp$, then output $\perp$. Otherwise, output $m = i - 1$.

For the verification, output the result of $\text{GVerify}(\text{gpk}, \Sigma, M)$ where $C = (\Sigma, M)$. Note that, since gsigk and gsigk' are originally sampled, the **Open** algorithm does not output 1 (resp. 2) when it identifies that gsigk (resp. gsigk') was used. Here, without loss of generality, we implicitly assume that the **Open** algorithm outputs 1 (resp. 2) when it identifies that gsigk (resp. gsigk') was used.

Next, we consider to provide a formal definition of public verifiability by following BUFF (Beyond UnForgeability Features) security [3, 15, 17, 33] where an adversary outputs a key for verification. Assume that pk contains a plaintext space which we denote $\text{MessageSpace}(\text{pk})$. Let an adversary of the public verifiability output a ciphertext $C^* = (\Sigma^*, M^*)$ and a public key pk^* such that C^*

is valid under gpk (i.e., $\text{GVerify}(\text{gpk}, \Sigma^*, M^*) = 1$), the decryption result is in $\text{MessageSpace}(\text{pk}^*)$, and $\text{MessageSpace}(\text{pk}) \cap \text{MessageSpace}(\text{pk}^*) = \emptyset$. This definition guarantees that the decryption result of a valid ciphertext under pk is in $\text{MessageSpace}(\text{pk})$. Moreover, producing such a pk^* violates the last winning condition of full traceability since pk^* contains a signing key of an uncorrupted user in the GS context. This, we can expect that public verifiability can be reduced to full traceability of the underlying GS scheme. However, there is an artificial construction: if the PKE.Dec algorithm outputs $m \notin \text{MessageSpace}(\text{pk})$, then outputs $\perp$ (or assigns a plaintext on $\text{MessageSpace}(\text{pk})$ and outputs this message). Then, there is no possibility that an adversary can produce C^* and pk^* where the decryption result is in $\text{MessageSpace}(\text{pk}^*)$, and $\text{MessageSpace}(\text{pk}) \cap \text{MessageSpace}(\text{pk}^*) = \emptyset$. We conclude that the definition is incomplete to capture what public verifiability can be realized when the underlying GS scheme provides full traceability.

We state the above incomplete definition as a stepping stone to explore what level of public verifiability can be achieved when the underlying GS scheme provides full traceability. Providing a formal definition is left as future work in this paper.

5.2 On Public Verifiability of PKE

Due to our observation in the previous section, we pointed out that previous definitions of public verifiability also can be bypassed by the artificial construction. In this section, we discuss these definitions given in [20, 21].

Nieto et al. [21] defined strictly non-trivial public verification for general encryption that contains PKE, identity-based encryption, and tag-based encryption. Because we focus on PKE in this paper, we remove IDSp and TagSp (in their notions) below. They introduced a parameter generation algorithm PG that takes a security parameter λ and outputs public parameters par , and the key generation algorithm PKE.KeyGen takes par as input. We follow the notion below. We also remark that the syntax of their verification algorithm differs from ours where the Ver algorithm outputs either $\perp$ if the ciphertext fails the validation or a (transformed) ciphertext. The decryption algorithm Dec' takes the transformed ciphertext (and the decryption key), and outputs a plaintext or $\perp$. The actual decryption algorithm (Dec in the following definition) is represented as the composition of the verification algorithm and the decryption algorithm, i.e., $\text{Dec} = \text{Dec}' \circ \text{Ver}$.

Definition 6 (Strictly Non-Trivial Public Verification [21]). *Let $\text{par} \leftarrow \text{PG}(1^\lambda)$. Ver is said to be strictly non-trivial if, for all $(\text{pk}, \text{dk}) \leftarrow \text{PKE.KeyGen}(\text{par})$,*

Condition 1. $\text{Ver}(\text{par}, \text{pk}, C) = 0 \iff \text{Dec}(\text{par}, \text{pk}, \text{dk}, C) = \perp$ for all C .

Condition 2. *There exists a ciphertext C for which $\text{Dec}(\text{par}, \text{pk}, \text{dk}, C) = \perp$.*

They mentioned that “*Condition 1 requires that successful public verification is both necessary and sufficient for the decryption algorithm not to fail.*”. Moreover, they mentioned that “*Condition 2 formally excludes IND-CCA-secure schemes where Dec never outputs $\perp$ (e.g. [2, 37, 38] where modified (challenge) ciphertexts decrypt to random messages) to capture the intuition that in order to determine whether C carries some meaningful message one must have at least partial knowledge of the private key (which contradicts the goals of strictly non-trivial public verification).*”. However, still we can consider an artificial construction. For any IND-CCA secure PKE scheme where there is a ciphertext for which the decryption result is $\perp$, the key generation algorithm chooses an invalid ciphertext, say C_{invalid} , and contains it to the public key. For a ciphertext C , the verification algorithm Ver outputs 0 only when C_{invalid} is input, and outputs C otherwise. If the decryption algorithm Dec' takes 0, then it outputs $\perp$. Otherwise, now Dec' takes C (and the decryption

key) as input. If the decryption algorithm of the underlying PKE scheme returns $\perp$, then Dec' outputs a random non- $\perp$ plaintext (it can be pre-defined when Dec' is a deterministic algorithm). Otherwise, if the decryption algorithm of the underlying PKE scheme does not return $\perp$, then Dec' outputs the decryption result. This artificial construction does not affect the correctness since honestly generated ciphertext is correctly decrypted due to the correctness of the underlying PKE scheme. Moreover, it provides strictly non-trivial public verification. Here, what we argue in this paper is that the definition of strictly non-trivial public verification *by itself* can be bypassed by an artificial construction, and we do not claim that the above artificial construction does not affect the IND-CCA security. As a final remark, Nieto et al. mentioned that “*all IND-CCA-secure PKE schemes trivially achieve public verifiability with respect to $\text{Ver}(\text{par}, \text{pk}, C) := C$ and $\text{Dec}' := \text{Dec}$. We are often interested in the case where something non-trivial is occurring in Ver , i.e. where the consistency check is essential for successful decryption.*”⁶ We totally agree with the concept.

Faonio, Hofheinz, and Russo [20] also defined public verifiability in a replayable CCA (RCCA) context.

Definition 7 (Public Verifiability [20]). *PKE is a public key scheme with publicly verifiable ciphertexts if there is a deterministic algorithm Ver which, on input (pk, C) outputs an error symbol $\perp$ whenever $\text{Dec}(\text{pk}, \text{dk}, C) = \perp$, else it outputs valid.*

This also can be bypassed by modifying the decryption algorithm where if a decryption result is $\perp$, then the algorithm outputs a (pre-defined) non- $\perp$ plaintext, and the Ver algorithm always outputs valid. Again, what we argue in this paper is that the definition of public verifiability *by itself* can be bypassed by an artificial construction, and we do not claim that the above artificial construction does not affect the RCCA security.

6 Conclusion

In this paper, we explore the winning conditions of full traceability. We show that the condition, where an adversary is required to produce a valid group signature whose opening result is an uncorrupted user, is essential. As a by-product of our result, we also explore public verifiability of a GS-based PKE scheme and state that providing a formal definition is left as future work in this paper.

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⁶Here, we replace “generic encryption” to “PKE” and remove a tag t from the input of Ver from their sentences.

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